GATE CS — Engineering Mathematics

Concise study notes for GATE 2027 · built from the 2016–2026 papers in this repo

Where the marks come from (average marks per paper, 2016–2026)

Topic	Marks / paper	Priority
Probability & Statistics	3.3	Tier 1
Linear Algebra	2.8	Tier 1
Sets, Relations, Groups, Lattices	2.3	Tier 2
Graph Theory	2.3	Tier 2
Combinatorics & Recurrences	1.5	Tier 2
Propositional & First-Order Logic	1.3	Tier 3
Calculus	1.3	Tier 3

Engineering Mathematics carries about **14.8 of the 85 technical marks** in a typical paper, the largest share of any subject. It is scoring because questions are short and mostly numerical.

Contents

1. Probability & Statistics
2. Linear Algebra
3. Sets, Relations, Functions, Groups, Lattices
4. Graph Theory
5. Combinatorics, Recurrences, Generating Functions
6. Propositional & First-Order Logic
7. Calculus
8. Exam strategy & revision checklist

Legend: **Trap** = mistake that cost marks in past papers · **Tip** = shortcut. For practice, use `GATE_CS_Engineering_Maths.pdf` (past questions + new questions with solutions).

1. Probability & Statistics

Basic rules

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ · $P(A^c) = 1 - P(A)$
Conditional: $P(A|B) = P(A \cap B) / P(B)$
Independent: $P(A \cap B) = P(A) \cdot P(B)$ (then $P(A|B) = P(A)$)
Total probability: $P(A) = \Sigma\, P(A|B_i) \cdot P(B_i)$ over a partition $\{B_i\}$
Bayes: $P(B_i|A) = P(A|B_i)P(B_i) / \Sigma_j\, P(A|B_j)P(B_j)$

> **Trap:** mutually exclusive events with non-zero probabilities are *dependent* ($P(A \cap B)=0 \neq P(A)P(B)$). "At least one" → compute 1 − P(none).

Random variables

- $E[X] = \Sigma\, x \cdot p(x)$ or $\int x\, f(x)dx$. Linearity always holds: $E[aX+bY] = aE[X]+bE[Y]$, even if X, Y are dependent.
- $Var(X) = E[X^2] - (E[X])^2$. $Var(aX+b) = a^2 Var(X)$.
- $Var(X+Y) = Var\, X + Var\, Y + 2Cov(X,Y)$; $Cov(X,Y) = E[XY] - E[X]E[Y]$.
- Independent ⇒ uncorrelated (Cov = 0), but not the other way round. $E[XY] = E[X]E[Y]$ if independent.
- Cauchy–Schwarz: $(Cov(X,Y))^2 \leq Var(X) \cdot Var(Y)$, so correlation $\rho \in [-1, 1]$.
- PDF: $f(x) \geq 0$ and $\int f = 1$ (use this to find the constant C). $P(a \leq X \leq b) = \int_a^b f$. CDF $F(x) = P(X \leq x)$.
- Median m: $F(m) = 0.5$. Mode: peak of pmf/pdf. For a symmetric distribution mean = median.

Standard distributions

Distribution	pmf / pdf	Mean	Variance
Bernoulli(p)	$P(1)=p$, $P(0)=q=1-p$	p	pq
Binomial(n,p)	$C(n,k)p^k q^{n-k}$	np	npq
Geometric(p) (trials until first success)	$q^{k-1}p$, $k \geq 1$	1/p	q/p^2
Poisson(λ)	$e^{-\lambda}\lambda^k/k!$	λ	λ
Uniform(a,b)	$1/(b-a)$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$\lambda e^{-\lambda x}$, $x \geq 0$	$1/\lambda$	$1/\lambda^2$
Normal(μ,σ²)	bell curve, symmetric about μ	μ	σ^2

- Exponential is **memoryless**: $P(X > s+t \mid X > s) = P(X > t) = e^{-\lambda t}$.
- Normal: about 68% within 1σ, 95% within 2σ, 99.7% within 3σ. Standardise $Z = (X-\mu)/\sigma$. The pdf $(1/(\sigma\sqrt{2\pi})) \cdot \exp(-(x-\mu)^2/2\sigma^2)$ identifies μ and σ directly (e.g. $\exp(-x^2/18)$ → $\sigma^2 = 9$).
- Poisson: $E[X^2] = \lambda + \lambda^2$, $E[X(X-1)] = \lambda^2$. Example: mean 5 ⇒ $E[(X+2)^2] = Var + (E[X+2])^2 = 5 + 49 = 54$.
- Sum of independent Poissons is Poisson with the λs added; same for independent Normals (means and variances add).

Counting-style probability tricks

- **Indicator variables:** write X as a sum of 0/1 variables and use linearity. Expected number of fixed points of a random permutation = 1. Expected number of elements in a random subset etc.
- **Symmetry:** in a random permutation, P(1 comes before 2) = 1/2.

- **Pólya urn** (start 1 red + 1 blue, add one ball of the drawn colour each step): after n steps the number of red balls is uniform on {1, …, n+1}. So P(equal after 2 steps) = 1/3.
- **Random bits:** for non-zero $a \in \{0,1\}^n$ and uniform x, P(a·x is odd) = 1/2.
- **Coupon collector:** expected draws to see all n coupons = $n \cdot H_n = n(1 + 1/2 + \ldots + 1/n)$.
- Two dice: sum ways for 2…12 are 1,2,3,4,5,6,5,4,3,2,1 (out of 36). All-distinct on 6 dice: $6!/6^6 = 5/324$.
- Hypothesis/diagnosis problems → draw a tree and apply Bayes. Expect a table of 4–6 numbers.

2. Linear Algebra

Determinants

$\det(AB) = \det A \cdot \det B \cdot \det(A^T) = \det A \cdot \det(A^{-1}) = 1/\det A$
$\det(kA) = k^n \det A$ for an $n \times n$ matrix (e.g. 4×4, $\det 3 \to \det(2A) = 16 \cdot 3 = 48$)
Swapping two rows flips the sign; adding a multiple of one row to another changes nothing; a zero row/column or two equal rows $\Rightarrow \det = 0$
Triangular/diagonal matrix: det = product of diagonal entries
$A \cdot \text{adj}(A) = \det(A) \cdot I \cdot \det(\text{adj } A) = (\det A)^{n-1} \cdot A^{-1} = \text{adj}(A)/\det(A)$ (exists iff $\det \neq 0$)

Rank and linear systems

- Rank = number of non-zero rows in echelon form = max number of independent rows (= columns). $\text{rank}(A) \leq \min(m,n)$; $\text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B)$.
- Rank–nullity: $\text{rank}(A) + \text{nullity}(A) =$ number of columns.
- For $Ax = b$ with n unknowns: consistent iff $\text{rank}(A) = \text{rank}([A|b])$. Unique solution if that rank $= n$; infinitely many if $< n$; none if ranks differ.
- Homogeneous $Ax = 0$: always has $x = 0$. A non-trivial solution exists iff rank $< n$ (for square A, iff $\det A = 0$).
- Square A: invertible $\Leftrightarrow \det \neq 0 \Leftrightarrow$ full rank $\Leftrightarrow$ no zero eigenvalue $\Leftrightarrow$ columns independent.
- Rank of the $n \times n$ all-ones matrix is 1. A vector in the null space of M means $Mv = 0$, so M has eigenvalue 0 (M is singular).

Eigenvalues and eigenvectors

$Av = \lambda v$, $v \neq 0$ · solve $\det(A - \lambda I) = 0$ · for 2×2: $\lambda^2 - (\text{trace})\lambda + \det = 0$
$\Sigma\lambda_i = \text{trace}(A)$ · $\Pi\lambda_i = \det(A)$

Matrix	Eigenvalues
A^k	λ^k
A^{-1}	$1/\lambda$
$A + cI$	$\lambda + c$
cA	$c\lambda$
A^T	same as A
Triangular / diagonal	the diagonal entries
Real symmetric	all real; eigenvectors of distinct eigenvalues are orthogonal
Idempotent ($A^2=A$)	0 or 1
Nilpotent ($A^k=0$)	all 0
Orthogonal	$\lvert\lambda\rvert = 1$
$n \times n$ all-ones	n (once), 0 ($n-1$ times)

- Eigenvectors for distinct eigenvalues are linearly independent; n distinct eigenvalues $\Rightarrow$ diagonalisable.
- Geometric multiplicity $\leq$ algebraic multiplicity. The maximum multiplicity of an eigenvalue of an $n \times n$ matrix is n (e.g. the identity).
- **Cayley–Hamilton:** A satisfies its own characteristic equation. Use it to compute A^{-1} or high powers.
- Singular $\Leftrightarrow$ 0 is an eigenvalue.

> **Trap:** trace and determinant give two equations; with 3 unknown eigenvalues you need a third piece of information (e.g. rank or a given eigenvalue).

LU decomposition

- A = LU, L lower triangular (unit diagonal in Doolittle), U upper triangular. Exists without row swaps iff all leading principal minors are non-zero.
- Use to solve Ax = b: Ly = b (forward substitution), then Ux = y (back substitution). Cost $O(n^3)$ to factor, $O(n^2)$ per solve. det A = product of U's diagonal (when L is unit).

3. Sets, Relations, Functions, Groups, Lattices

Counting relations on a set of n elements

Type	Count
All relations	$2^{(n^2)}$
Reflexive (or irreflexive)	$2^{(n^2-n)}$
Symmetric	$2^{(n(n+1)/2)}$
Reflexive and symmetric	$2^{(n(n-1)/2)}$
Antisymmetric	$2^n \cdot 3^{(n(n-1)/2)}$
Asymmetric	$3^{(n(n-1)/2)}$
Equivalence relations	Bell number: 1, 2, 5, 15, 52, 203 (n = 1…6)
Partial orders (labelled)	1, 3, 19, 219, 4231 (n = 1…5)
Total orders	n!

- Equivalence = reflexive + symmetric + transitive; it partitions the set into equivalence classes.
- Partial order = reflexive + antisymmetric + transitive. Strict order = irreflexive + transitive.
- Transitive closure by Warshall's algorithm. Composition: $(a,c) \in S \circ R$ if $\exists b$ with $(a,b) \in R$ and $(b,c) \in S$.

Functions ($|A| = m$, $|B| = n$)

- All functions $A \to B$: n^m · one-to-one: $n!/(n-m)!$ $(m \le n)$ · bijections: $n!$ $(m = n)$.
- Onto: $\Sigma_k (-1)^k C(n,k)(n-k)^m = n! \cdot S(m,n)$, using Stirling numbers.
- $|P(A)| = 2^m$ · $|A \times B| = mn$ · $|A \cup B| = |A| + |B| - |A \cap B|$.
- $f \circ g$ is injective $\Rightarrow$ g is injective; $f \circ g$ is surjective $\Rightarrow$ f is surjective.

Partial orders and lattices

- Hasse diagram: drop self-loops, transitive edges, and arrow directions. Edge = covering relation.
- **Maximal/minimal** elements may be many; **greatest/least** must be unique and comparable with all.
- Chain = all elements comparable; antichain = none comparable. **Dilworth:** minimum number of chains covering the poset = size of the largest antichain.
- **Lattice:** every pair has a least upper bound (join) and greatest lower bound (meet). Every finite lattice is bounded.
- Divisibility on D_n (divisors of n): join = lcm, meet = gcd. D_n is a lattice; it is a Boolean algebra iff n is square-free. D_n with $n = p_1^{a_1} \dots p_k^{a_k}$ is a product of chains (grid); its Hasse diagram has $\Sigma_i a_i \cdot \Pi_{j \neq i}(a_j+1)$ edges (D_{36}: 12).
- Subset lattice $(P(S), \subseteq)$: join = $\cup$, meet = $\cap$, a Boolean algebra with 2^n elements.
- Distributive lattice $\Leftrightarrow$ it has no sublattice isomorphic to M_3 or N_5. Complemented: every element has a complement. Boolean algebra = complemented distributive lattice.

Algebraic structures

Structure	Requirements
Semigroup	closure, associativity
Monoid	semigroup + identity
Group	monoid + inverses
Abelian group	group + commutativity

- **Lagrange:** the order of a subgroup divides |G|. The order of every element divides |G|. A group of prime order is cyclic (and abelian).
- Cyclic group of order n: generators = $\varphi(n)$ (Euler totient); for each divisor d there is exactly one subgroup of order d, so the number of subgroups is the number of divisors of n; $\varphi(d)$ elements have order d.
- Groups of order 4: Z_4 and $Z_2 \times Z_2$. Order 6: Z_6 and S_3 (non-abelian). Order p^2: always abelian.
- $(Z_n, +_n)$ is a group. $(Z_n^*, \times_n)$ (units mod n) is a group of order $\varphi(n)$; $(Z_p^*, \times)$ is cyclic for prime p.
- Subgroup test: non-empty and $a \cdot b^{-1} \in H$ for all $a, b \in H$. A subset of a finite group closed under the operation is a subgroup.
- S_n (permutations) has n! elements; non-abelian for $n \geq 3$.

> **Trap:** “Is (S, op) a group?” — check closure first, then identity, then inverse for *every* element (e.g. integers under multiplication fail on inverses; zero has no inverse under multiplication).

4. Graph Theory

Basics

- **Handshake lemma:** Σ deg(v) = 2E. The number of odd-degree vertices is even. In a directed graph Σ in-degree = Σ out-degree = E.
- Simple graph: max edges n(n−1)/2 (K_n). Labelled graphs on n vertices: 2^(n(n−1)/2). $K_{m,n}$ has mn edges.
- Tree: connected and acyclic; n−1 edges; unique path between any two vertices; at least 2 leaves. Binary tree: $n_0 = n_2 + 1$ (leaves = degree-2 nodes + 1).
- Connected graph: E ≥ n−1. With c components: E ≥ n−c. A disconnected simple graph has at most (n−1)(n−2)/2 edges, so (n−1)(n−2)/2 + 1 edges force connectivity.
- Complement graph: E(G) + E(G^c) = n(n−1)/2. A graph is bipartite iff it has no odd cycle.

Counting spanning trees

- K_n: n^{n-2} (Cayley). Cycle C_n: n. Tree: 1. $K_{m,n}$: $m^{n-1} \cdot n^{m-1}$.
- General: **Kirchhoff's matrix-tree theorem**: any cofactor of the Laplacian L = D − A.
- MST: unique if all edge weights are distinct. Squaring positive weights keeps the same MST (order is unchanged) but can change shortest paths.

Planarity

Euler (connected planar): V − E + F = 2
E ≤ 3V − 6 (V ≥ 3) · for triangle-free planar graphs E ≤ 2V − 4
Sum of face degrees = 2E · K_5 and $K_{3,3}$ are non-planar (Kuratowski: a graph is planar iff it has no subdivision of K_5 or $K_{3,3}$)

Euler and Hamiltonian

- Euler circuit: connected and every vertex has even degree. Euler path (not circuit): exactly two odd-degree vertices.
- Hamiltonian: no easy iff condition. Sufficient: Dirac (every deg ≥ n/2), Ore (deg u + deg v ≥ n for non-adjacent pairs). K_n (n ≥ 3) has (n−1)!/2 Hamiltonian cycles.

Colouring

- χ(K_n) = n · χ(bipartite) = 2 · χ(C_n) = 2 if n even, 3 if n odd · χ(tree) = 2 · χ(planar) ≤ 4 · χ ≥ clique size.
- Brooks: χ ≤ Δ unless G is complete or an odd cycle (then Δ+1). Always χ ≤ Δ + 1.
- Chromatic polynomial: K_n → k(k−1)…(k−n+1); tree with n vertices → $k(k-1)^{n-1}$; cycle C_n → $(k-1)^n + (-1)^n(k-1)$.

Matching, independence, cover

- Perfect matching in K_{2n}: (2n)!/(2^n·n!). In $K_{n,n}$: n!.
- **Hall's theorem:** a bipartite graph has a matching saturating X iff |N(S)| ≥ |S| for every S ⊆ X.
- **König:** in a bipartite graph, maximum matching = minimum vertex cover.
- Independent set size + vertex cover size = n (in the same graph, for max/min versions).
- Triangle-free graph on n vertices has at most ⌊$n^2/4$⌋ edges (Mantel).
- Isomorphism invariants: number of vertices, edges, degree sequence, cycle lengths, connectedness. Equal invariants do not prove isomorphism.

5. Combinatorics, Recurrences, Generating Functions

Counting toolbox

Permutations $P(n,r) = n!/(n-r)!$ · Combinations $C(n,r) = n!/(r!(n-r)!)$
Circular arrangements of n objects: $(n-1)!$ (divide by 2 more if flipping allowed)
Arrangements of a word with repeats: $n!/(n_1!n_2!\ldots)$
Stars and bars: non-negative solutions of $x_1+\ldots+x_k = n$: $C(n+k-1, k-1)$. With $x_i \geq 1$ for all i: $C(n-1, k-1)$.

- For lower bounds $x_i \geq a_i$, shift: $y_i = x_i - a_i$. For upper bounds use **inclusion–exclusion** (subtract solutions where a variable exceeds its bound).
- Inclusion–exclusion: $|A_1 \cup \ldots \cup A_n| = \Sigma|A_i| - \Sigma|A_i \cap A_j| + \Sigma|A_i \cap A_j \cap A_k| - \ldots$
- **Pigeonhole:** n+1 objects into n boxes ⇒ some box has ≥ 2. In general some box has ≥ $\lceil m/n \rceil$.
- Binomial: $(x+y)^n = \Sigma C(n,k)x^k y^{n-k}$ · $\Sigma C(n,k) = 2^n$ · $\Sigma(-1)^k C(n,k) = 0$ · $C(n,k) = C(n,n-k)$ · $C(n,k)+C(n,k+1) = C(n+1,k+1)$ · Vandermonde: $\Sigma\, C(m,i)C(n,k-i) = C(m+n,k)$.
- Lattice paths from (0,0) to (m,n) with right/up steps: $C(m+n, n)$.

Special numbers

Name	Meaning	Values
Derangements D_n	permutations with no fixed point; $D_n = n!\Sigma(-1)^k/k!$; $D_n = (n-1)(D_{n-1}+D_{n-2})$	0, 1, 2, 9, 44, 265 (n = 1…6)
Catalan $C_n = C(2n,n)/(n+1)$	binary trees / BSTs with n nodes, balanced parentheses, stack-permutations	1, 1, 2, 5, 14, 42, 132
Fibonacci	binary strings of length n with no two consecutive 1s = F_{n+2}	1, 1, 2, 3, 5, 8, 13, 21
Stirling S(n,k)	partitions of n objects into k non-empty blocks; $S(n,k) = kS(n-1,k) + S(n-1,k-1)$	S(4,2)=7, S(4,3)=6
Bell B_n	all partitions = $\Sigma_k\, S(n,k)$	1, 2, 5, 15, 52, 203

Recurrence relations

- **Linear homogeneous, constant coefficients:** $a_n = c_1 a_{n-1} + c_2 a_{n-2}$. Characteristic equation $r^2 - c_1 r - c_2 = 0$. Distinct roots r_1, r_2: $a_n = A\, r_1^n + B\, r_2^n$. Repeated root r: $a_n = (A + Bn)r^n$. Use initial conditions to find A, B.
- Non-homogeneous: a_n = homogeneous solution + particular solution (guess same form as the right-hand side; if it clashes with a root, multiply by n).
- Common results: $T(n) = T(n-1) + n \to n(n+1)/2$ · $T(n) = 2T(n-1) + 1 \to 2^n - 1$ (Tower of Hanoi) · $T(n) = T(n-1) + c \to cn$ · $T(n) = 2T(n/2) + n \to n \log n$.
- Ways to climb n steps with 1 or 2 steps: Fibonacci. With 1, 2, 3 steps: $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ (1, 1, 2, 4, 7, 13, 24, 44, 81, 149, 274).

Generating functions

$1/(1-x) = \Sigma x^n$ · $1/(1-ax) = \Sigma a^n x^n$ · $(1+x)^n = \Sigma C(n,k)x^k$ · $e^x = \Sigma x^n/n!$
$1/(1-x)^k = \Sigma C(n+k-1, k-1)x^n$ · $x/(1-x)^2 = \Sigma n \cdot x^n$

- Product of generating functions = convolution of the sequences. Multiplying by $1/(1-x)$ gives partial sums. Multiplying by x^k shifts the sequence right by k.
- Solve a recurrence: multiply by x^n, sum, solve for G(x), expand by partial fractions.
- Counting with GF: choose from each “kind” with allowed counts, multiply the per-kind

polynomials, read off the coefficient of x^n.

6. Propositional & First-Order Logic

Propositional logic

p → q ≡ ¬p ∨ q ≡ ¬q → ¬p (contrapositive) · converse q → p and inverse ¬p → ¬q are **not** equivalent to p → q
p ↔ q ≡ (p→q)∧(q→p) ≡ ¬(p⊕q) · De Morgan: ¬(p∧q) ≡ ¬p∨¬q, ¬(p∨q) ≡ ¬p∧¬q
¬(p → q) ≡ p ∧ ¬q · Absorption: p ∨ (p∧q) ≡ p · p → (q → r) ≡ (p∧q) → r

- Tautology: always true. Contradiction: always false. Satisfiable: true for some assignment. n variables ⇒ 2^n rows; 2^(2^n) distinct Boolean functions.
- Falsify an implication with p = T, q = F — a fast way to test validity of formulas built from →.
- Rules of inference: modus ponens (p, p→q ⊢ q), modus tollens (¬q, p→q ⊢ ¬p), hypothetical syllogism (p→q, q→r ⊢ p→r), disjunctive syllogism (p∨q, ¬p ⊢ q), resolution (p∨q, ¬p∨r ⊢ q∨r).
- Functionally complete sets: {¬,∧}, {¬,∨}, {¬,→}, {NAND}, {NOR}. {∧,∨} alone is not complete.
- Normal forms: CNF = AND of ORs; DNF = OR of ANDs. SAT is NP-complete; Horn-SAT is polynomial.

> **Trap:** “p only if q” means p → q; “p if q” means q → p; “p unless q” means ¬q → p. Affirming the consequent (q, p→q ⊢ p) is invalid.

First-order logic

- Translation: “All P are Q” → ∀x (P(x) → Q(x)). “Some P is Q” → ∃x (P(x) ∧ Q(x)). “No P is Q” → ∀x (P(x) → ¬Q(x)).
- Negation: ¬∀x φ ≡ ∃x ¬φ and ¬∃x φ ≡ ∀x ¬φ.
- ∀ distributes over ∧: ∀x(P∧Q) ≡ ∀xP ∧ ∀xQ. ∃ distributes over ∨. But ∀x(P∨Q) does not imply ∀xP ∨ ∀xQ, and although ∃x(P∧Q) ⇒ ∃xP ∧ ∃xQ holds, the converse does not.
- **Quantifier order:** ∀x∀y and ∃x∃y can be swapped. ∃x∀y φ ⇒ ∀y∃x φ always; the converse fails (counter-example: R(x,y) is “x = y” on a 2-element domain).
- Free vs bound variables; a sentence has none free. Prenex normal form: all quantifiers at the front.
- Validity: true in every interpretation. FOL validity is semi-decidable (undecidable in general).

> **Trap:** ∀x(P(x) ∧ Q(x)) is much stronger than “all P are Q”; ∃x(P(x) → Q(x)) is trivially true whenever some x is not P. Always use → with ∀ and ∧ with ∃.

7. Calculus

Limits

lim sin x / x = 1 · lim (1 − cos x)/x² = 1/2 · lim (eˣ − 1)/x = 1 · lim ln(1+x)/x = 1 · lim tan x / x = 1
lim (1 + 1/n)ⁿ = e · lim (1 + a/n)ⁿ = eᵃ · lim (1+x)^(1/x) = e (x → 0) · lim xⁿ/eˣ = 0 as x → ∞

- **L'Hôpital:** for 0/0 or ∞/∞, lim f/g = lim f′/g′ (repeat if needed).
- **Maclaurin series** make limits fast: eˣ = 1 + x + x²/2! + …, sin x = x − x³/3! + …, cos x = 1 − x²/2! + …, ln(1+x) = x − x²/2 + x³/3 − …, 1/(1−x) = 1 + x + x² + …
- Rational function as x → ∞: compare degrees (equal → ratio of leading coefficients; numerator lower → 0).
- Squeeze theorem: g ≤ f ≤ h and lim g = lim h = L ⇒ lim f = L.

Continuity and differentiability

- Continuous at a: $\lim_{x\to a^-} = \lim_{x\to a^+} = f(a)$. Differentiable ⇒ continuous; the converse is false (|x| at 0).
- Piecewise functions: check left/right limits and left/right derivatives at the joins. Sum of $|x - a_i|$ terms is non-differentiable exactly at the points a_i.
- Rules: (uv)′ = u′v + uv′ · (u/v)′ = (u′v − uv′)/v² · chain rule (f∘g)′ = f′(g)·g′.

Maxima and minima; theorems

- Critical points: f′(x) = 0. f″ > 0 → local min; f″ < 0 → local max; f″ = 0 → inconclusive. On a closed interval [a,b] compare critical points **and endpoints**.
- max/min of a sin x + b cos x is ±√(a²+b²).
- **Rolle:** f continuous on [a,b], differentiable on (a,b), f(a) = f(b) ⇒ f′(c) = 0 for some c. **Mean value theorem:** f(b) − f(a) = f′(c)(b − a). **IVT:** a continuous function takes every value between f(a) and f(b). **Extreme value theorem:** a continuous function on [a,b] attains its max and min.

Integration

- ∫xⁿdx = x^(n+1)/(n+1), ∫dx/x = ln|x|, ∫eˣ = eˣ, ∫sin = −cos, ∫cos = sin. By parts: ∫u dv = uv − ∫v du.
- Even f: $\int_{-a}^{a} f = 2\int_0^a f$. Odd f: $\int_{-a}^{a} f = 0$. Also $\int_0^a f(x)dx = \int_0^a f(a-x)dx$; for symmetric integrands like sin/(sin+cos) on [0, π/2] the answer is half the interval: π/4.
- Gamma/Wallis: ∫₀^∞ xⁿe^(−x) dx = n! · ∫₀^∞ e^(−x²) dx = √π/2 · ∫₀^(π/2) sin²x dx = π/4.
- Improper: ∫₁^∞ dx/xᵖ converges iff p > 1; ∫₀¹ dx/xᵖ converges iff p < 1.
- Leibniz: d/dx ∫ₐ^(g(x)) f(t)dt = f(g(x))·g′(x).
- Series: Σ1/nᵖ converges iff p > 1; geometric Σrⁿ = 1/(1−r) for |r| < 1; telescoping Σ1/(n(n+1)) = 1; ratio test: converges if lim |a_(n+1)/aₙ| < 1.

8. Exam Strategy & Revision Checklist

Study order (by marks per hour)

1. **Probability** — Bayes, expectation/variance, distributions table, indicator-variable tricks.
2. **Linear algebra** — determinant and eigenvalue properties, rank and consistency of linear systems.
3. **Sets, relations, groups, lattices** and **graph theory** — formula-driven; memorise the tables above.
4. **Combinatorics** — stars and bars, inclusion–exclusion, recurrence solving, Catalan/Fibonacci/derangements.
5. **Logic** and **calculus** — fewer marks, but fast once the equivalences and standard limits are known.

Exam-day habits

- Many Engineering Maths questions are NAT with no negative marking: always attempt them. MCQs lose ⅓ (1-mark) or ⅔ (2-mark) marks when wrong.
- Test small cases ($n = 2, 3$) to check a formula before choosing an option.
- For “which statements are TRUE” (MSQ), prove or find a counter-example for each statement separately; no partial marks.
- Do arithmetic twice in numerical problems; round exactly as asked (2 or 3 decimals).

Checklist before the exam

- ☐ Write the distribution table (mean, variance) from memory.
- ☐ State the determinant and eigenvalue properties from memory.
- ☐ Recite relation-counting formulas, Lagrange, and the lattice definitions.
- ☐ Recall Euler's formula, $E \leq 3V - 6$, Cayley, Hall, König, χ values.
- ☐ Compute stars-and-bars with bounds without looking.
- ☐ Convert English to FOL with correct $\forall/\rightarrow$ and $\exists/\wedge$ pairing.
- ☐ Standard limits, Maclaurin series, MVT/Rolle conditions.
- ☐ Redo all past questions in `GATE_CS_Engineering_Maths.pdf` and check them against the official answer keys.